

The Stretch-Shortening Cycle: Proposed Mechanisms and Methods for Enhancement

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SUMMARY

EFFICIENT STRETCH-SHORTENING CYCLE (SSC) MECHANICS RESULT IN ENERGY CONSERVATION AND ENHANCED PROPULSIVE FORCES. BASED ON THE CURRENT REVIEW OF RESEARCH, ELASTIC ENERGY THROUGH TENDON RECOIL AND AN INCREASE IN ACTIVE STATE BECAUSE OF AN INCREASE IN THE WORKING RANGE SEEM THE MOST PLAUSIBLE EXPLANATION FOR THE SSC MECHANISM. USING THE RESEARCH PRESENTED, THE SSC MECHANISM MAY BE BEST DEVELOPED USING THE PLYOMETRICS PYRAMID ENSURING THAT THE ATHLETE IS TECHNICALLY COMPETENT AT EACH STAGE BEFORE PROGRESSING IN INTENSITY AND COMPLEXITY. STRENGTH TRAINING AND SSC TRAINING SHOULD BE PERFORMED CONCURRENTLY. SSC DRILLS SHOULD BE BASED ON STRENGTH CAPACITIES AND SPORT-SPECIFIC VARIABLES.

INTRODUCTION

It is well established that a vertical jump preceded by a countermovement (i.e., a prestretch) will increase vertical displacement above a squat jump (one with no prestretch) (10).

Investigations have revealed improvements in the range of 18–20% (15) to 20–30% (13) and a difference in maximum jump height of approximately 2–4 cm (9). Moreover, by increasing the load applied and the rate of loading during the countermovement, for example, after a run-up or a depth jump, jump height may further increase (4,8,77,78). This phenomenon is a consequence of what is termed the stretch-shortening cycle (SSC), which describes an eccentric phase or stretch followed by an isometric transitional period (amortization phase), leading into an explosive concentric action. The SSC is synonymous with plyometrics (37) and is often referred to as the reversible action of muscles (112). Other examples of SSC actions include the natural parts of movements such as running or walking or the windup movement in throwing.

Aside from an enhanced concentric contraction (propulsive force), efficient usage of the SSC also affords the athlete with a reduction in the metabolic cost of movement (9,10). This may be evidenced with data suggesting that the energetic cost of running for animals with heavy limbs is about the same as those with light limbs (as heavier limbs would increase the load applied and the rate of loading) (40). In addition, Verkhoshansky (101) and Voigt et al. (102) reported that

economical sprinting (i.e., efficient usage of the SSC) can recover approximately 60% of the total mechanical energy (40% being replenished by metabolic processes during the following cycle). In addition, the contribution of nonmetabolic energy sources increase with increases in running velocity (24,101,102).

The SSC is therefore essential to many sporting movements, with performance dependent on its efficient use within a movement skill. Consequently, many coaches look to incorporate training drills such as plyometrics, which can enhance the athlete's use of the SSC (68,77,85,91,93,95,97). The successful integration of these exercises, however, may only be achieved with an understanding of the underpinning mechanics. Several mechanisms have been proposed to explain the SSC phenomenon, with varying emphasis and conclusions reached across the literature. Investigators have reported the significance of elastic strain energy (11,33,64,74,77), involuntary nervous processes (12,13), increased active range of movement (9,10), length-tension characteristics (30,34), preactivity tension

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(68,95), and enhanced coordination because of the innate action of the prestretch (9,10). The purpose of this article, therefore, is to review the most current theories to discuss and define the processes that underpin the SSC. A second aim is to identify training strategies to enhance the SSC processes to optimize performance.

STORAGE OF ELASTIC ENERGY

During hopping, jumping, and running, for example, our legs exhibit similar characteristics to a spring, whereby the leg spring compresses on ground contact and stores energy, before rebounding at push-off and releasing energy (50). It is currently recognized that the tendon is the primary site for the storage of elastic energy (EE) (64,74). The magnitude of stored EE (often referred to as strain or potential energy) is hypothesized to be proportional to the applied force and the induced deformation (112). Previous research supports that elasticity plays a substantial role in enhancing the motor output in sport movements (13,33,64,74,77) and likely explains the 20–30% difference seen between a countermovement jump (CMJ) and a squat jump (SJ) (13). In addition, Verkhoshansky (101) reported a high correlation ($r = 0.785$) between the tendon's capacity to store EE and the performance of distance runners.

Although recognizing the fundamental role of EE, some investigators do not support its significant role in enhancing force production (9,10). Instead, they place emphasis on its ability to reduce the metabolic cost of movement. For example, Bobbert et al. (10) suggests that if propulsive displacement is equal between 2 athletes, the athlete who most optimally uses SSC mechanics would incur less metabolic work. They argue that lengthening of the series elastic component (SEC), namely, the tendon, occurs at the expense of length over which the contractile elements can produce force. This is in agreement with Bobbert and Casius (9) who explained that despite the SEC contributing to maximum jump height through storage of EE, it does not

explain the differences in jump height among various types of jumps. In summary, therefore, an inverse relationship may exist between EE release and force production via the contractile components. Put simply, the greater the release of EE, the greater the reduction in cross-bridge formation and concomitant force output from these structures. Research by Lichtwark and Wilson (74) may also provide support for this theory. Their research suggests that muscles act at high values of efficiency by contracting fibers at favorable speeds, which are often different from the speed of the whole muscle-tendon complex (MTC). Therefore, through the use of EE, some fibers are deactivated during periods of shortening. They do suggest, however, that during maximal efforts, EE can enhance force production. The correlation reported above by Verkhoshansky (101), therefore, maybe because of the fact that improved performance in long-distance events with respect to the SEC is because of its ability to conserve energy and therefore provide an efficient energy release/conservation system as opposed to one that increases force per stride. Although this does seem plausible, many investigators still cite that EE can indeed improve force output. Perhaps, however, this occurs during maximal rather than submaximal force outputs when energy conservation is not a priority. For example, the SEC may actually increase force per stride during the 100-m sprint or increase competitive jump displacement in the long jump or high jump. In addition, many authors report that the contractile components, when not lengthening or shortening, are in fact undergoing an isometric action, which is optimal for force generation during SSC activities (39,63,105). The role of EE appears contentious and will be discussed further in this article along with additional mechanisms that may further explain the increase in power output after an SSC.

It should be noted that there is a point of diminishing returns whereby once the eccentric loading (stretch phase)

reaches a critical threshold, the subsequent concentric contraction exhibits no further increase in force/EE return and may even result in reduced force output. This is likely because the change from the eccentric contraction into the propulsive concentric contraction (i.e., amortization phase) takes too long (106). This may be a consequence of involuntary neural inhibition (discussed later in this article) ultimately causing the EE to be released and lost as heat energy during the amortization phase (69). In addition, Wilson et al. (106) found that the SSC had a half-life of 0.85 seconds and that by 1 second, the benefits diminished by 55%. This investigation, however, examined the countermovement within a bench press (of male weightlifters) and thus results may be more indicative of upper-body mechanics. Therefore, if EE is to be optimally used, the load experienced within the eccentric phase should be within the limits of the athlete and the amortization phase should be as rapid as possible. It is well recognized that these variables differ among athletes and, most importantly, are measurable and trainable (37,87). This can further be exemplified by reviews from Newton and Dugan (87) and Flanagan and Comyns (37) who reported that untrained individuals generally attain higher vertical displacements after the CMJ compared with drop jumps. As the individual's SSC mechanics develop, however, this reverses and higher values are achieved during drop jumps. Moreover, with progressive training, the athlete can continue to increase the drop height with concomitant increases in jump height.

MECHANICAL MODEL

To further understand the role of EE, it is important to consider the mechanical model of muscle function devised by Hill (49). Hill (49) suggests that force can be analyzed as the summation of 3 components: a contractile component, namely, actin and myosin; a parallel elastic component (PEC), comprising the sarcolemma and muscle fascia; and a SEC,

comprising the cross-bridges, structural proteins, and tendons. The PEC is responsible for force exerted by a relaxed (passive) muscle when it is stretched beyond its resting length. Its contribution of stored mechanical energy is considered small and consequently so too is its contribution to propulsive force. Conversely, the SEC is put under tension by the force developed in actively contracted muscles. Because an actively contracted muscle resists stretching, particularly if the stretching is imposed rapidly (28,29,63), the SEC results in considerable storage of energy.

Most research describes the tendon as the workhorse of the SEC (64,74); however, the additional components have still been credited with significance. The following sections thus provide a brief review of the components of the SEC that may contribute to the enhanced power output after the SSC.

CONTRACTILE FILAMENTS AND STRUCTURAL PROTEINS

The time span for cross-bridge maintenance has been estimated to be in the region of 15–120 milliseconds (96). The amortization period therefore needs to be minimal to augment energy return from these structures (96) because energy is lost at the instance of detachment (37). To the contrary, Fleck and Kraemer (37) hypothesize that the period of amortization is simply too long to allow any significant contribution from cross-bridge maintenance, which they estimate to be in the region of 30 milliseconds. They therefore suggest that alternate mechanisms are responsible for the SSC phenomenon.

Bosco et al. (14), however, suggests that there is a difference in cross-bridge life times between fast-twitch (FT) and slow-twitch (ST) fibers, suggesting that they exhibit different viscoelastic properties. In agreement, Siff (96) explains that muscles that are rich in FT fibers would benefit from a rapid short-range SSC. Conversely, slower larger amplitude jumps with a longer transient period would benefit muscles rich in ST fibers. Schmidtbleicher (94)

describes these as short and long SSC, respectively, whereby the former has a ground contact time (GCT) of <250 milliseconds (e.g., a drop jump) and the latter >250 ms (e.g., a CMJ).

It has also been suggested that the prestretch of active muscles alters the properties of the contractile machinery (10), whereby cross-bridges may become “stuck” on stretch and release slower than when followed by an isometric or concentric action (72). In addition, the force produced by tetanized single muscle fibers may be augmented by a prestretch (28,29). This enhancement has been shown to increase with the speed of prestretch and to decrease with the amount of time elapsed after the prestretch (28,29,63).

Finally, the structural protein titin, which spans half the sarcomere from the M-line to the Z disc, has been proposed as an explanation of passive force enhancement (72). It has been suggested that titin attaches to actin in a calcium-dependent manner so that its length becomes smaller and stiffness increases on muscle activation when calcium is released from the sarcoplasmic reticulum into the sarcoplasm (69,72). The role of titin within the SSC mechanics is relatively novel and as such requires further research.

TENDONS

Tendons are considered the key site for energy storage within the SEC because of their ability to extend and store energy and recoil and release energy (64,74). Kubo et al. (64) suggests that the EE stored in tendons is a key mechanism underpinning the SSC phenomenon. This is in agreement with Lichtwark and Wilson (74) who suggest that tendon recoil is responsible for both increasing power output and conserving energy during locomotion. The elastic properties of a tendon are therefore very important for power production and efficiency.

In contrast to the tendon, muscle tissue is not efficient at energy storage and return. However, because muscle and tendon are arranged in series, they are

subjected to the same forces (112). The distribution of stored energy among these tissues therefore is dependent on their deformation, which in turn is a function of stiffness or its inverse value compliance (112). Put simply, whichever tissue structure stretches the most will store the most EE. For example, during passive stretch, the stiffness of the PEC is less than 100 times that of the tendon and thus the majority of deformation occurs in the PEC (71). Conversely, during active movement, the stiffness of the muscle tissue and its surrounding PEC far exceeds that of the tendon, therefore reversing the storage site for EE (110).

As a consequence of the tendon's superior ability to store and release energy, a goal of all athlete training programs should be the optimal transfer of potential energy arising from a prestretch being delivered to these structures. This transfer, however, can only be optimized through the development of muscle stiffness throughout the prestretch. Stiffness and compliance are therefore key terminology when explaining the efficiency of the SSC and the enhanced power output noted in trained athletes. The following sections thus provide an overview of the stiffness-compliance continuum and the performance-related benefits of muscle stiffness.

STIFFNESS AND COMPLIANCE

Zatsiorsky and Kraemer (112) explain that while the stiffness of a tendon is constant, the stiffness of a muscle is variable and depends on the forces exerted (i.e., the muscle is compliant when passive and stiff when active). Through training, particularly plyometrics (68,77,85,91,93,95,97), it is possible to develop high forces and maintain high levels of stiffness in muscles, exceeding that of tendons. In such a scenario, whereby the muscle does not stretch, the tendon is forced to. As discussed above, this is optimal with elite athletes demonstrating a superior ability to store EE primarily in their tendons (2,24,46,50).

Leg stiffness can be defined as the ratio of maximal ground reaction force to maximum leg compression during the mid stance phase (50) or by dividing the change in force by the change in length (55). Komi (58) suggests that higher stiffness levels of lower limb muscles during SSC exercises increase the amount of stored and reused EE. Also, Bojsen-Moller et al. (11) found that power, force, and velocity parameters obtained during jumping had a significant and positive correlation to tendon stiffness. In agreement, leg stiffness has been shown to augment with an increase in jump height and hopping height (3,31,32), and knee joint stiffness, that is, the ability to resist flexion, has been shown to be crucial to performance after a drop jump (53). Arampatzis et al. (3) also demonstrated that GCT and ankle joint stiffness were inversely related during drop jumps. Similarly, Kuitunen et al. (67) also demonstrated that subjects with the highest stiffness in the ankle joint had the shortest GCT at all running speeds and that these times are also related to decreased flexion in joints such as the knee and hip. Furthermore, a positive correlation exists between rate of force development (RFD) and connective tissue stiffness in the lower body (11) and the upper body (106). Heise and Martin (46) and Dalleau et al. (24) reported a further positive correlation between leg stiffness and running economy, concluding that economical runners possessed a running style that was stiffer during ground contact. Therefore, in agreement with Wilson and Flanagan (105), there appears to be a strong relationship between the amount of stiffness in a human system and various parameters of sports performance.

DEVELOPING MUSCLE STIFFNESS

Leg stiffness largely depends on ankle stiffness (3,31,32), and joint stiffness in general depends on antagonistic co-contraction (17,54,89). These in turn are regulated by muscle tension at landing (preactivation; 50) and the concerted actions of the involuntary reflexive neural processes (discussed later in this article). For example, co-

contraction between the plantarflexor and dorsiflexor muscles and the knee extensor and knee flexor muscles will increase joint stiffness throughout the whole leg in preparation for ground impact (54). In agreement, Kuitunen et al. (67) found that as running speed increased, preactivation of the plantar flexors and knee extensors increased, increasing MTC stiffness and the ability to tolerate and absorb high-impact loads at the beginning of the contact phase. Moreover, the preactivation of the triceps surae muscle coupled with the stretch reflex and Golgi tendon organ (GTO) inhibition will ensure high muscular stiffness at ground contact to support and propel the body (59). Finally, Gollhofer et al. (43) and McBride et al. (77) found that increase in preactivation and eccentric phase muscle activity increased concentric work output.

In a study by Hobara et al. (50), it was shown that power-trained athletes (e.g., sprinters) show higher leg stiffness and ankle stiffness than endurance-trained athletes and untrained individuals. Furthermore, these athletes exhibit less GCT and longer aerial time during hopping. This is in agreement with Arampatzis et al. (2) who also reported that, as well as exhibiting higher maximal plantar flexion moments, sprinters also have higher stiffness in the triceps surae tendon and that a significant relationship exists between the two ($r = 0.817$). The higher tendon stiffness is likely to reduce the probability of tendon strain injuries (excess elongation may lead to partial or complete tendon ruptures), which would occur as a consequence of stronger muscles (2,83). In addition, the increased tendon and muscle stiffness may increase joint stability through its resistance to joint displacement (16). Therefore, it may be concluded that both muscle and tendon tissue show plasticity to sufficient external mechanical loads. Both undergo hypertrophy and share the characteristic of enhanced collagen synthesis resulting in stiffer muscles and also stiffer tendons. It may be deduced, however, that this increased

tendon stiffness reduces elongation and impairs elastic strain energy. In support, investigations by Kubo et al. (64) reveal that tendon stiffness in the knee extensors is inversely correlated with countermovement prestretch augmentation. This is conceivable because prestretch augmentation has been shown to be significantly greater in individuals with compliant tendon structures because this allows for greater EE storage under a given extent of force (18,64,83). Conversely, with a stiff tendon, all the shortening must occur in the muscle tissue, which under the high velocities of the SSC is not optimal for contractile force (74).

In summary, despite stiffness possibly increasing in both tissues (7), the ratio of muscle to tendon stiffness increase may be such that optimal efficiency is maintained based on anatomical location and continued imposed demands. For example, Lichtwark and Wilson (74) suggest that tendon stiffness is optimal to achieve the highest efficiency in both walking and running. Also, the human Achilles tendon can strain up to 10.3% during 1-legged hopping (73), whereas the anterior tibialis tendon has a maximum recorded value of 3.1% (75). In addition, Markovic and Jaric (76) also explain that the muscular system is adapted to provide maximum mechanical output under accustomed loads of daily living. Therefore long-term exposure to higher or lower loads would then result in shifting the optimal loading for maximal mechanical output. Similarly, it has also been reported that the stiffness of the lower limbs may be limited to prevent injury, whereby high stiffness may increase stress to the anatomical structures during ground contact phases (16,24,80).

In support of the research by Markovic and Jaric (76), human tendon stiffness has been shown to increase only after resistance training using heavy loads (61,66,92) and isometric squats (61,66). Therefore, any increase in muscle strength would be offset by stiffer tendons (65). Conversely, jumping, sprint training (86,99), and drop jump

protocols (62) have produced no significant changes in tendon properties. This led investigators to conclude that an exercise protocol centered on ballistic contractions (rapid acceleration against resistance) cannot change the tendon properties and that adaptations leading to increases in joint stiffness are therefore related to the significant changes in mechanical properties of the muscle (e.g., active cross-bridges) not the tendon (65). Siff (96) also hypothesized that increases in strength and stiffness are achieved with increased loading and increased rate of loading. Therefore, training should emphasize exercises with high acceleration methods. Taken collectively, the research may allow for the following deduction. Strength training should precede plyometric training to develop sufficient strength in the muscles and tendons and reduce the probability of tendon injuries. After strength training, however, plyometrics (and ballistics alike) must be performed to create a more favorable environment for structural adaptations in the muscle only, whereby increases in the muscle to tendon strength ratio are noted and stiffness becomes optimal for force production/maintenance. The above literature summarized the role of EE in the increased propulsive force noted after SSC actions. The following sections focus on additional mechanisms, namely, the neurophysiological model, active state development, isometric muscle actions, and length-tension characteristics.

NEUROPHYSIOLOGICAL MODEL

It has been suggested that the muscle spindle may be responsible for the potentiation after a prestretch because of its initiate reflex recruitment of additional motor units or increased rate of firing of already recruited units (12,16). This mechanism may partly contribute to the development of a high level of active state, enabling the muscles to generate larger forces and thus more work during the concentric phase before the start of push-off (10). Electromyographic (EMG) analysis, however, does not

support this hypothesis. Many studies have reported a nonsignificant change in EMG activity after a prestretch when compared with a non-prestretch action (33,65,98). It may be concluded, therefore, that reflex activity does not account for the increased force caused by the SSC (9,10,65). It is also interesting to note that myoelectric activity does not change during maximal vertical jump performance with or without additional loads (96). This suggests that ballistic actions maximally activate motor units regardless of muscle shortening velocity and force production during the concentric phase (33,65,77,98,103).

Neural reflexes, however, despite possibly not explaining the difference in vertical displacements of various jumps, are likely intimately involved in the final power output after SSC-type activities. The nervous stimulation to the muscle during the eccentric phase of an SSC is modified by the concerted actions of the muscle spindle and GTO (68,95). Zatsiorsky and Kraemer (112) summarized this relationship with the following example: during landing after a drop jump, the stretch applied to the leg extensor causes the muscle spindle to contract that muscle. However, the sudden high muscular tension causes the GTO to simultaneously inhibit its activity. This drop in muscle tension prevents the muscle and tendon from incurring damage. Therefore, if athletes are not accustomed to these exercises and loads, the activity of the extensor muscles during takeoff is inhibited by the GTO. In support, Schmidtbleicher et al. (95) reported that in subjects unaccustomed to intense SSC movements, EMG activity was reduced from 50 to 100 milliseconds before ground contact and lasted for 100–200 milliseconds. After plyometric training, however, it is possible to reduce the inhibitory effects (disinhibition) of the GTO, whereby the athlete is able to sustain high landing forces without a decrease in exerted muscular force (68,95). The intensity (e.g., dropping height or load) may then be increased.

For example, Kyrolainen et al. (68) reported that after 4 months of SSC training, the preactivity of muscles increased and this change led to increased MTC stiffness.

When this data are examined alongside the above theories underpinning the efficacy of muscle stiffness, additional conclusions can be proposed. When an athlete is trying to generate maximal muscular effort and attain high levels of muscle stiffness, there becomes a trade-off between the 2 reflexes and volitional muscle activation. It is evident that it is not simply a case of the athlete contracting as hard as possible. Ultimately, the final power output is regulated by subconscious neural reflexes. Moreover, the intensity of each reflex, which is not constant, determines the final outcome (112). Therefore, the objective of drop jumps, for example, may be to expose the athletes to fast muscle stretching rather than to immediately generate large forces (112). Although nervous reflexes may not explain the potentiation after the SSC, they may be able to limit it. As suggested above, however, it may take up to 4 months of plyometric training to inhibit the GTO and enable the potentiation of the muscle spindle, if indeed it does occur.

WORKING RANGE

Investigations by Bobbert et al. (10) and Bobbert and Casius (9) and a comprehensive review by Van Ingen et al. (100) reported that the greater jump height seen in the CMJ compared with the SJ was exclusively because of the fact that greater active state could be developed during the prestretch. This results in an increased impulse (force $\times$ time) and thus a greater change in vertical velocity of the body. They suggested that this may be because of the greater moments occurring at the hip, knee, and ankle joints, enabling the production of greater force and work over the first part of the propulsive phase. This also led to an increase in vertical velocity over the entire concentric range (100) and greater ground reaction force at

the start of push-off in the CMJ compared with other jumps (10).

The time delay in reaching peak force is in part because of the finite rate of increase in muscle stimulation by the central nervous system, the propagation of the action potential on the muscle membrane, time constants of calcium release and cross-bridge formation, and the interaction between contractile elements and the SEC (10,84,111). In addition, the slack, caused by the crimped orientation of tendon fibers (71) (described by the toe region of the tendon force-deformation curve), must be stretched out of the SEC before it will transmit any force to the skeletal system (49). This factor is reportedly a significant contributor to the time to develop peak force (111). Collectively, this delay in reactivity is known as the electromechanical delay (EMD) and describes the interval between the time of onset of muscle activity and the time of onset of mechanical output (19,84). This time delay is consequential to commencing movement from zero to low muscular tension, and its negative effects can be reduced by enabling the muscle to build up a maximum active state before the start of the propulsive phase through either an isometric contraction (e.g., preloading) or a counter-movement (10). Moreover, it has been hypothesized that increases in MTC stiffness would decrease the EMD, allowing muscles to generate tension more rapidly and counteract deleterious forces at joints (105).

In support of an active range, Bobbert et al. (10) found a direct relationship ($r = 0.88$) between the time to stimulate the gluteus maximus and vertical ground reaction force in a non-counter-movement jumping task. In addition, with the use of a simulated spring model, Bobbert and Casius (9) further demonstrated that by increasing the rate of muscular stimulation during an SJ, the difference in vertical ground reaction force was reduced because this in turn reduced the distance covered at a submaximal active state. They therefore concluded that the difference

between CMJ and SJ heights would vary among individuals depending on their ability to develop force. A small difference would be seen in those who could develop force quickly, whereas a large difference would be seen in those who developed forces at slower rates. The authors attributed this to the faster cross-bridge cycling rates of FT fibers and hence the ability to build maximal force at greater rates than ST fibers. To the contrary, however, Bosco et al. (14) found no difference in jump heights after short SSCs between individuals with predominantly FT or ST muscle fiber composition. However, and in agreement with Van Ingen et al. (100), the investigators reported that individuals with predominantly ST muscle fibers benefited most from longer SSCs such as the CMJ. Finni et al. (33) provided further contention to the above theories by reporting that participants jumped higher when using a CMJ compared with a condition that allowed for the build up of maximal isometric force before the vertical jump. They therefore concluded that other factors, such as EE, must contribute to the enhanced performance after a prestretch.

Research supporting the significance of the development of active state, however, may gain further credibility when examined concurrently with research concerning the time available to develop peak force (i.e., RFD). Aagaard (1) hypothesized that in skilled athletes, this takes between of 0.25 and 0.4 seconds, with force linearly increasing throughout. Other researchers, however, suggest that maximum force development may require 0.6 to 0.8 seconds (27,58). A prestretch therefore provides additional time over which force can be developed, ensuring that by the time of the concentric contraction, the accumulated force is above that of the starting force of the SJ, leading to more powerful propulsion. Fundamental to coaches, RFD is trainable, with advanced athletes from power-orientated sports reaching peak values quicker (112) and perhaps explaining their increase in jump heights.

MUSCLE-TENDON INTERRELATIONSHIP AND ISOMETRIC CONTRACTIONS

Kubo et al. (60) examined the MTC of the human medial gastrocnemius during ankle dorsiflexion to plantarflexion. The movement consisted of dorsiflexion at 2 different speeds of lengthening, followed by plantarflexion. The investigators revealed that the tendon length increased to a greater extent in the fast lengthening condition. No significant change was found in the fascicle length during the first half of plantarflexion, whereas the tendon rapidly shortened. Both the tendon and muscle rapidly shortened in the second half of plantarflexion, suggesting that during the transition from prestretch to shortening, the muscle contracted isometrically. These findings are in agreement with Fukunaga et al. (38) who found that during the stance phase of walking, the medial gastrocnemius muscle contracted isometrically, whereas the tendon lengthened by 7 mm. During the push-off phase, however, both the tendon and muscle rapidly shortened. The isometric action may be of benefit because it likely avoids the lowered force output that occurs with increasing velocity and can also far exceed the force output of concentric contractions (27). Fukunaga et al. (38) and Wilson and Flanagan (105) hypothesized that before the isometric action, muscles independently lengthen toward their optimal length-tension relationship to increase concentric force output. This is in agreement with Lichtwark and Wilson (74) who suggested that muscles contract at favorable speeds, which may differ from the MTC. In further support, Ishikawa et al. (56) found that the medial gastrocnemius only lengthened during the early stance phase of walking, whereas the soleus continued to lengthen until the end of the stance phase, when both muscles rapidly shortened at toe-off.

LENGTH-TENSION RELATIONSHIP

As mentioned above, muscle or fascicle length may be another mechanism involved in increasing force output after an SSC. Finni et al. (33) demonstrated

that the vastus lateralis generates more force with a prestretch compared with no prestretch, yet there is no difference in EMG activity (as discussed above). The force enhancement may therefore be related to a longer muscle or fascicle length before the concentric phase, placing the muscle in a more advantageous position on the length-tension diagram to produce force (36). In agreement, Ettema et al. (30) explained that during the start of propulsion in both the CMJ and SJ, the muscle fibers are on the descending limb of their length-tension relationship. However, in the CMJ, because of stretching of the SEC, they are less beyond their optimal length and are therefore able to produce greater force over the first part of their concentric range.

METHODS TO ENHANCE SSC MECHANISMS

Because the SSC clearly plays an important role in performance in many sports, then developing this capacity via effective training practices is crucial. Reportedly, the optimal method to train SSC motor skills is plyometrics (68,77,85,91,93,95,97). The following sections outline how plyometric exercises can be progressively integrated into an athlete's training program and also outline appropriate methods of performance evaluation. The practical suggestions herein will be made based on the evidence for the SSC mechanisms highlighted in the preceding sections.

Plyometrics cover a wide range of jumping, hopping, and bounding-based exercises that have the fundamental aim of enhancing SSC performance. Although appearing relatively simple tasks, for example, a CMJ or a drop jump, plyometric exercises are in fact very complex and fundamental movement skills. As such, appropriate time should be allocated to the development of these skills, and the strength and conditioning coach should ensure that the athlete displays mastery in these before progressing to additional drills. This requires a progressive system of exercises to be set up, through which an athlete can pass to ensure that they have

the required technical mastery to be able to perform the entire gamut of plyometric exercises in a fashion that both maximizes performance gains and also minimizes injury risk.

Ideally, in terms of maximizing performance, plyometric training should be preceded by strength training to reduce the risk of injury to the MTC and increase the quality and quantity of type II fibers. The latter point is of significance because of the high correlation between the percentage of type II fibers and peak power output (23) and is therefore likely to increase the athletes' net potential to develop power (58). As a consequence of the size principle of motor unit recruitment (47), strength training, that is, $\geq 85\%$ 1 repetition maximum (RM), ≤ 6 repetitions, 2–6 sets, 2- to 5-minute rest (6), is required to recruit these type II fibers (45). Although this sequence of strength training preceding plyometric training is undoubtedly physiologically sound, it may not optimize sequential development based on a motor skill basis, and holding back the introduction of plyometric development until a sound base of strength has been developed may not maximize long-term development. The reality of modern sports is that performers will compete in their sports at a very young age. Sports, such as basketball, football, soccer, and the like, inherently contain a large number of SSC activities and involve numerous jumping and landing activities. These activities are more often than not undertaken before an adequate strength base has been established, and therefore, the development of effective landing techniques, for example, takes on a very important role. Additionally, and as described above, plyometric exercises contain a large skill component in addition to a physical component. It therefore seems logical to progressively develop both components in a concurrent fashion, rather than having to develop the skill capacities from scratch once the strength base has been established. It is, however, prudent to ensure that plyometric exercises are prescribed based on

the athlete's progression of physical capacities.

INTRODUCING PLYOMETRIC EXERCISES—THE PLYOMETRIC PYRAMID

In introducing any skill, there needs to be a sequence of progression that allows an athlete to master basic components before moving onto more advanced exercises. Developing a progressive system requires a basic knowledge of the factors that determine plyometric intensity. Armed with this knowledge, exercises can be sequenced to provide appropriate stimulus for an athlete based on the aim of the program and on their physical and skill capacities at any time. Jeffreys (57) lists the determinants of plyometric intensity as follows:

1. The speed of movement, the greater the speed the greater the intensity.
2. The points of contact, with single-leg drills being more intense than double-leg drills.
3. The amplitude of movement, with greater amplitudes ground contact forces and hence increasing intensity.
4. The athlete's weight (or additional load), with the greater weight leading to higher intensities.

Additionally, exercises where an athlete moves from an eccentric movement to a concentric movement (e.g., a depth jump) are more intense than an eccentric movement alone (e.g., a drop land), other things being equal.

In essence, plyometric "skills" revolve around 2 basic capacities, jumping and landing. Although these are basic skills, a failure to adequately develop these will hinder the optimal application of plyometric exercises, limit athletic development, and also expose the athlete to greater injury potential. Based on the evaluation of plyometric intensity and the need to develop jumping and landing skills, Jeffreys (57) advocated the use of the plyometric pyramid as a method of introducing plyometric exercises. This involves 3 categories of exercise, all of which have a given aim and are able to alter intensity within each stage. Throughout these stages,



Figure. Jump up to box (left), drop land (middle) step from box (right) (before drop land or drop jump). When performing these drills, coaches should ensure that the athlete uses the appropriate foot contact and displays the correct limb alignment (i.e., shoulders in line with the knees, helping to place the center of gravity over the body's base of support, and ensuring no valgus or varus movement at the knees. In addition, the shoulders should be pulled down and back with the hands to side, ready to react.).

the main focus is on technical development, thus ensuring that on completion of the process, athletes are adequately prepared for the full range of plyometric exercises.

STAGE 1: JUMP TO BOX

This stage develops basic jumping abilities and also, crucially, landing ability in a controlled environment. By excluding the time gravity has to act and by teaching landing technique to beginner athletes or athletes with current landing problems, landing forces can be minimized. Varying the height of the box can provide a challenge to the athlete's jumping ability, while still minimizing landing forces. Moving from a double-leg to a single-leg landing can further challenge the athlete's landing ability. The Figure (left) illustrates an athlete demonstrating effective landing technique when using the jump to box.

STAGE 2: JUMP AND STICK

This stage builds on the athlete's landing capacity developed in stage 1 and develops their ability to control eccentric forces. Initially, exercises in this stage can involve low-amplitude movements, but progression can be provided by increasing the amplitude of movement and by moving from double- to single-leg landings. As well, by further developing landing technique, this stage allows the athlete to adapt to high landing forces (eccentric loads) through learned GTO disinhibition. This stage, and the amplitudes within, should be dictated by the quality of the movement and not be progressed until the athlete can stick the landing with appropriate levels of control and with appropriate foot contact. Heel contact, for example, is suggestive of GTO inhibition and the athlete's inability to optimally store energy in the tendons, which is essential to the amortization phase

(and duration of) used in the subsequent stage (35). In addition, and described in the preceding text, this stage also requires the development of muscle stiffness through preactivation tensioning and antagonist co-contraction and may therefore take several weeks to develop (68). The Figure (middle) illustrates an athlete demonstrating effective landing technique during a drop land.

STAGE 3: SHORT-RESPONSE JUMPS

This stage begins the true plyometric training where the SSC is used to enhance subsequent concentric performance. Here, the athlete performs jump activities of initially low amplitude, where the aim is to minimize GCT, while maintaining effective landing mechanics and body control. Again, this stage should be progressed to involve greater amplitude of jumps and the utilization of single-leg

activities. Ankling drills provide a good example of a short-response jump. In addition, there is research suggesting that overall leg stiffness is correlated by ankle stiffness (2,31,32); therefore, ankling may provide a prudent starting point. The Appendix provides description of the ankling drill.

Athletes who have moved through these stages should have developed the required skills to enable them to use the full gamut of plyometric exercises. The appropriate exercise to elicit the required physical adaptations can then be progressively introduced into the program.

START WITH THE END IN MIND

When constructing plyometric programs, coaches need to be acutely aware of the fundamental aim of their training and the precise physical capacities they are trying to develop. This allows the programming variables to be appropriately applied to elicit specific training effects.

One key factor when considering appropriate plyometric drills is the GCT involved in the activity to which the drill is aimed at. Hennessy and Kilty (48) and Schmidtleicher (94) found low correlations between jump heights after a CMJ and a drop jump, suggesting that these SSC activities are measuring different movement characteristics. To this end, Schmidtleicher (94) categorized plyometric activities as either slow SSC (>250 milliseconds) or fast (<250 milliseconds) SSC, depending on their GCT.

As an example, therefore, a CMJ (slow SSC) may be more suitable to train the acceleration phase of the 100 m, as Plisk (87) hypothesized that the force exerted by the front leg during push-off is applied for >250 milliseconds in elite sprinters, with some investigators reporting ranges of 0.34–0.37 milliseconds (81,83). Similarly, ski jumping (60), shot putting (70), and a standing takeoff in platform diving (83) have GCT in excess of 250 milliseconds and would benefit from CMJ training or derivatives of (e.g., tuck jump, split-SJ, jump over barrier, and single-leg

progressions). Conversely, a drop jump (fast SSC) and its derivatives (e.g., multiple hurdle jumps, bounding and single-leg progressions) would be more suitable to train top speed sprinting (82) and the take-off phase of the long jump (109) and high jump (24) because these SSC actions have a GCT less than 250 milliseconds. To this end, the GCT and the type and direction of forces should guide plyometric choice.

During plyometric exercises, strength and conditioning coaches should place emphasis on the importance of maximizing jump height (where applicable) while minimizing GCT (105,107,108). Although this seems logical, it has important implications for appropriate progression within plyometric programs. For example, where fast SSC activities are required, the use of additional loads, increased height of jumps, and the like may bring about an undesired increase in GCT and hence change the underlying nature of the exercise. Therefore, care should be taken when progressing plyometric activities so as not to negatively affect the main aim of the exercise. This is also the case where an attempt to make plyometrics more “sport specific,” by including balls and the like could result in a degradation of the physical performance, thus negating the ultimate training aim of the activity, that is, enhancing SSC performance.

EVALUATING THE SSC MECHANISM

As GCT is an important variable in plyometric training prescription, monitoring of this important variable is important and can be achieved using training/testing equipment, such as contact mats and force plates, and is available in real time, possibly facilitating athlete motivation (34,86). Moreover, calculation of the reactive strength index (height jumped/GCT) during activities, such as drop jumps, can provide strength and conditioning coaches with a good indication of an athletes’ SSC ability (35,36,89,107,108, 110). This is usually tested over the following drop heights: 30, 45, 60, and 75 cm (89). As previously mentioned,

efficient SSC mechanics should result in greater jump heights from greater drop heights. An additional method for monitoring prestretch augmentation is described by McGuigan et al. (79) and Walshe et al. (110) who compared the CMJ with the SJ and used the following formula: % prestretch augmentation = $([CMJ - SJ] \times SJ^{-1}) \times 100$. Alternatively, reactive strength may simply be calculated as $CMJ - SJ$ height (107). Although monitoring the athlete’s training adaptations to plyometrics training is considered fundamental, the optimal method to do this still needs to be fully elucidated and may simply depend on the availability of specialist equipment.

PLYOMETRICS AS PART OF A TOTAL PERFORMANCE PLAN

As with all training modalities, plyometrics should not be performed in isolation and instead as part of a total performance program that includes multiple modalities. It is therefore advised that the strength and conditioning coach should include additional ballistic exercises (explosive resisted movements in which the body or object is subjected to full acceleration) such as weightlifting movements because these may enhance the athlete’s power output throughout the triple extension (of the hips, knees, and ankles) (51) inherent to lower-body SSC motor skills. These exercises are also advocated to enhance Rate of Force Development (RFD) (35,44,51) specifically within the first 200 milliseconds (43) of force production. This may therefore assist in the development of active state, which as discussed, may be a fundamental tenet of SSC activities (9,10,100). Moreover, these may be of additional significance because the vast majority of athletic SSC movements occur within 0.3 seconds (111). As previously mentioned, strength training will also play a part in maximizing SSC activity. Although further research still needs to be carried out into the optimal application of plyometrics, some research suggests that its concurrent combination with power/ballistic

training (combination method) may produce superior results across a wide variety of athletic performance variables requiring power and speed when compared with using either method in isolation (22,55,90).

Additionally, SSC activity may be affected by its prior contractile history, and this needs to be taken into account when planning programs. For example, Comyns et al. (20) examined the acute effects of 3 back squats performed using 65, 80, and 93% 1RM on the performance of a DJ to determine if an optimal resistive load exists for complex training (see the references Docherty et al. (25) and Ebben (26) for a review of complex training). Results showed that all resistive loads reduced ($p < 0.01$) flight time and that lifting at the 93% load resulted in an improvement ($p < 0.05$) in GCT and leg stiffness. These results may suggest that heavy lifting will enhance the fast SSC mechanism (possibly through postactivation potentiation) because of a stiffer leg spring action, which in turn may benefit performance. However, it should also be noted that although some activities may provide the potential to enhance SSC activity, others may decrease it. Magnusson (75), for example, reported that static stretches resulted in an acute reduction in muscle stiffness and therefore should be avoided in warm-up activities for sessions where SSC activity is involved.

QUALITY NOT QUANTITY

Comyns et al. (21) examined the effect of a maximal SSC fatigue workout on the performance of a DJ performed 15, 45, 120, and 300 seconds after fatigue. The results indicated that the fatigue workout significantly reduced flight time ($p < 0.001$) and peak force ($p < 0.01$) and increased GCT ($p < 0.05$) at the 15-second interval, suggesting that the efficiency of the SSC mechanism was reduced. However, the results also showed a potentiation effect at the 300-second interval because of a significant increase in peak force and leg stiffness ($p < 0.05$). Significant to the former research finding, the negative effects of fatigue on the SSC

mechanism have also been demonstrated after submaximal intensity workouts (41,42,52,58,88) and after completion of a marathon run (5). Therefore, as the SSC mechanism may be negatively influenced by fatigue, the quality of movements should always be a critical factor in assessing performance during a session and will help a coach gauge appropriate volumes of plyometric activities. This knowledge should also guide the application of plyometrics within the annual macrocycle, with the activity being most effective in cycles without excessive fatigue.

CONCLUSION

Efficient SSC mechanics result in energy conservation of locomotion and enhanced propulsive forces. This efficiency, however, is largely a consequence of an individual's ability to transfer all stretch to the tendon through maintenance of muscle stiffness. In turn, this can only be achieved with sufficient plyometric training, enabling GTO disinhibition and subsequent preactivation tensioning and concomitant antagonistic co-contraction. Reportedly, the optimal method to train SSC movement skills is plyometrics, and appropriate drills include drop lands, whereby the body adapts to high landing forces, and drop jumps, whereby the focus shifts to reducing the amortization phase and therefore the loss of EE. In addition, plyometric training should be preceded by strength training to reduce the risk of injury to the MTC and increase the quality and quantity of type II fibers. Finally, because of the significance of active state, it is suggested that athletes train RFD through the use of ballistics such as plyometrics and weightlifting.

The potentiation effects of the muscle spindle remain contentious because of insignificant EMG tracings. Based on the current review of research, EE via tendon recoil and an increase in active state because of an increase in the working range seems the most plausible causes of the increase in force seen after SSC actions. These findings are in agreement with Wilson and Flanagan

(104) who conducted a similar review. They further speculated that the development of active state predominated in enhancing force output in a long SSC, whereas a short SSC relies more heavily on the reuse of EE.



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APPENDIX: ANKLING

During ankling, the knees should remain straight as the athlete steps from one foot to the other. Throughout the swing phase, the foot should be dorsiflexed with the big toe pointing up toward the shin. At ground contact and the instant before, the muscles of the foot and ankle should forcefully contract (creating MTC stiffness) and propel the body forward as the ankle moves into plantar flexion. Only the ball of the foot should contact the ground and the initial foot strike should be directly below the body's center of gravity. This exercise should initially be performed on the spot and then progressed to moving forward. Ankling is an inherent part of jumping and running activities and may therefore be considered a fundamental plyometric exercise.